

GR8

THE CLIMB SPACE

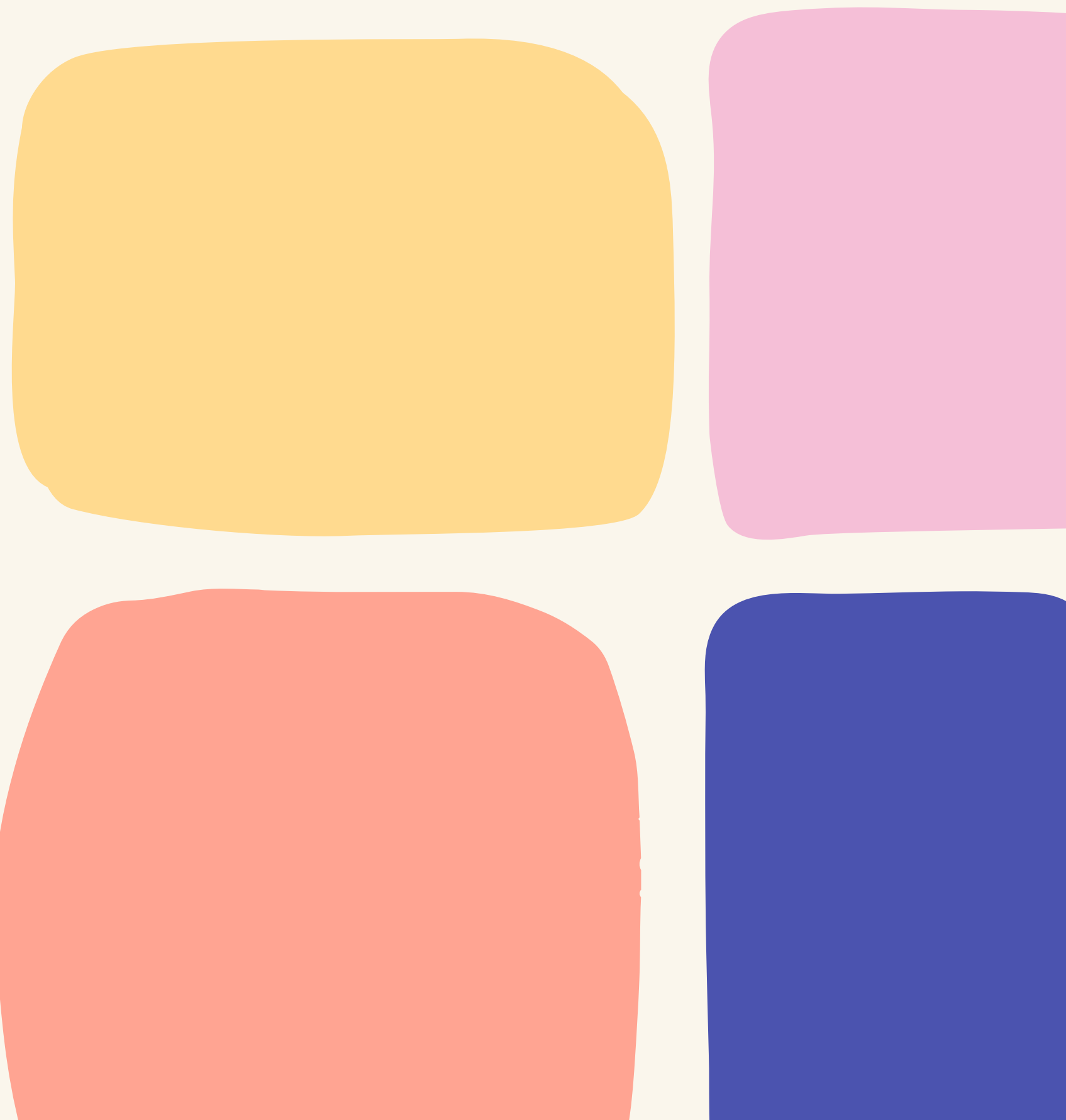
High-Fi Prototype

Team 18



ALTROUTE

Different bodies. Different Solutions.



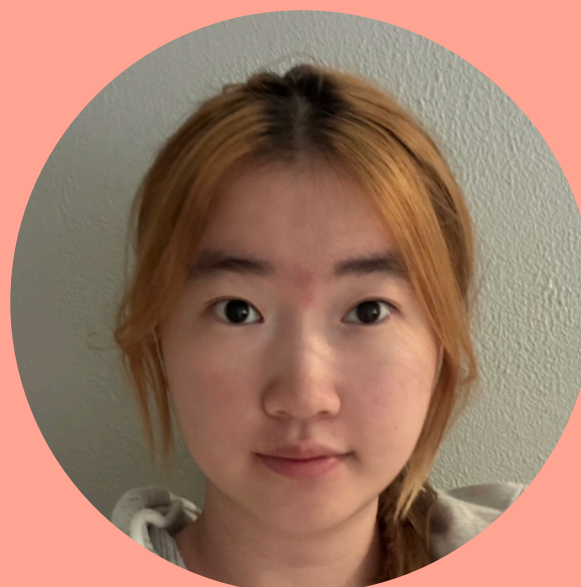
MEET THE TEAM



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Malaysia, KL

PROBLEM & SOLUTION OVERVIEW



WHAT'S THE PROBLEM?

Climbers often struggle to find community-shared betas that adequately accommodate their different body types and truly assist them in their climbs.

WHAT'S THE SOLUTION?

Provide personalized, AI-generated betas based on each climber's physiques and skill level, prioritizing their safety while maximizing satisfaction by still encouraging problem-solving

VALUES IN DESIGN



VALUES IN DESIGN

INCLUSIVITY

**USER
AGENCY**

ACCESSIBILITY

INCLUSIVITY

In-context meaning:

Climbing solutions should adapt to different bodies rather than assuming a single “correct” way to climb. The system supports diverse physiques, abilities, and constraints by generating personalized betas.

Physique-Aware Personalization

- Users input height, reach, skill level, and physical constraints
- AI generates multiple beta options tailored to body differences
- Removes reliance on “one-size-fits-all” community beta

Multiple Valid Solutions

- Several beta styles instead of one optimal path
- Encourages climbers to find strategies that fit their own movement style
- Frames difficulty as relational, not absolute

USER AGENCY

In-context meaning:

The climber controls how much assistance they receive, when to progress, and which strategy to follow. AI supports decision-making rather than replacing it.

Guidance Control

- Users control pacing (“Next,” “Repeat,” “Skip”)
- Users decide when to progress
- AI suggests — user chooses

Instruction Depth

- Gradual reveal instead of full solution upfront
- Maintains problem-solving and learning experience
- Supports both beginners and experienced climbers

Choice & Exploration

- Multiple beta options available for selection
- Users can switch strategies anytime
- AI suggests — user decides

ACCESSIBILITY

In-context meaning:

The system minimizes barriers to participation by using devices climbers already own and workflows that fit naturally into gym environments.

Low Barrier Hardware

- Designed for smartphones and optional earbuds
- No expensive or specialized equipment required
- Works within existing climbing routines

Flexible Use Modes

- Voice guidance optional
- Visual interaction still fully supported
- Users choose interaction style based on comfort and context

Seamless Gym Integration

- Quick interaction between attempts
- Hands-free audio option reduces need to touch phone with chalked hands
- Does not obstruct vision like AR headsets

OUTLINE

1. HEURISTIC EVALUATION RESULTS

2. UI AND PRODUCT REVISIONS

3. PROTOTYPE IMPLEMENTATION STATUS

4. PROTOTYPE

HEURISTIC EVALUATION RESULTS



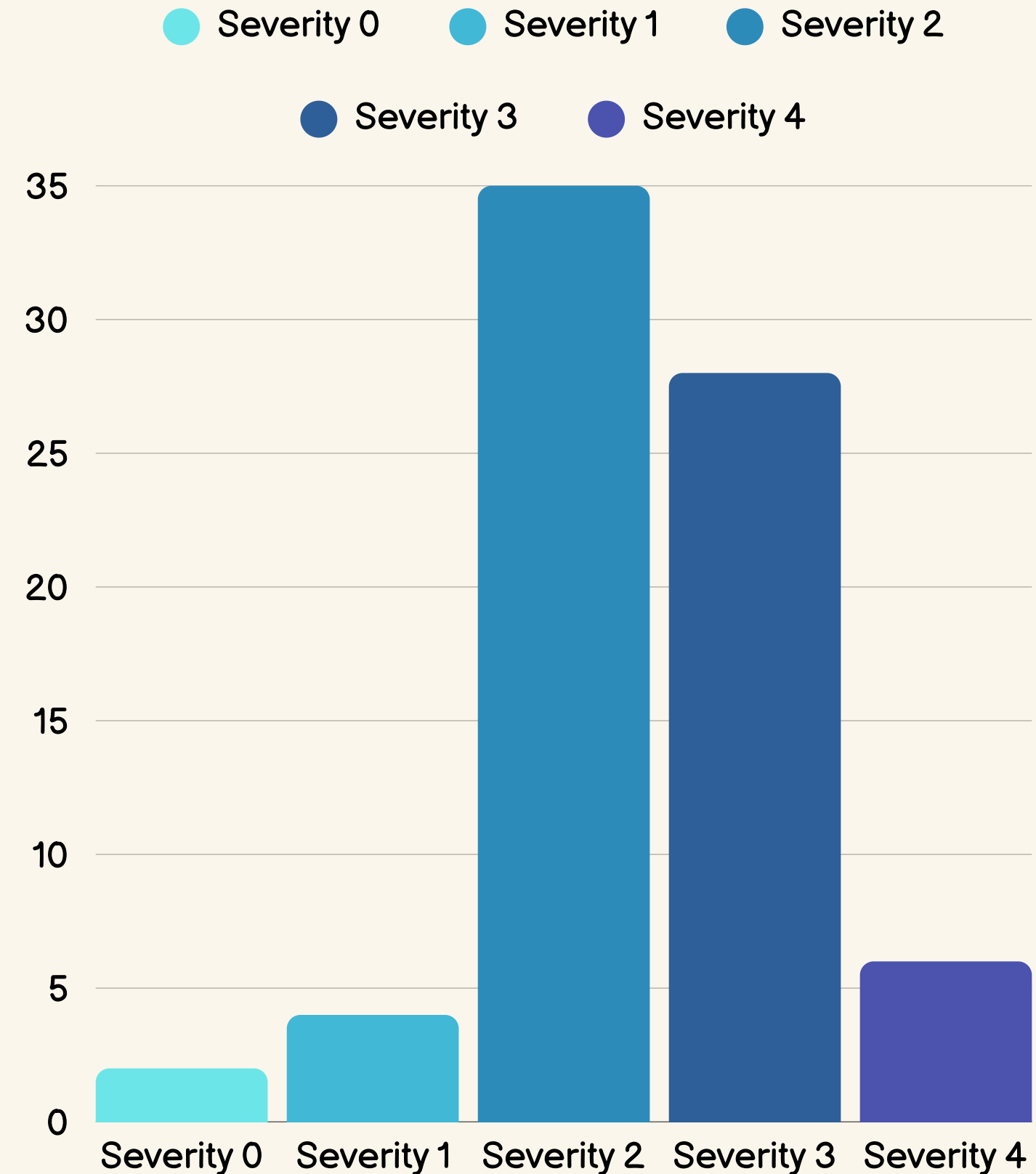
TOTAL VIOLATIONS

Heuristic	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	sum
Sev4			3	1	1				1		6
Sev3	4	3	4	2	4	2	2	2	3	2	28
Sev2	3	4		5	3	4	5	4	1	6	35
Sev1								4			4
Sev0										2	2
sum	7	7	7	8	8	6	7	10	5	10	75

VIOLATIONS BY SEVERITY

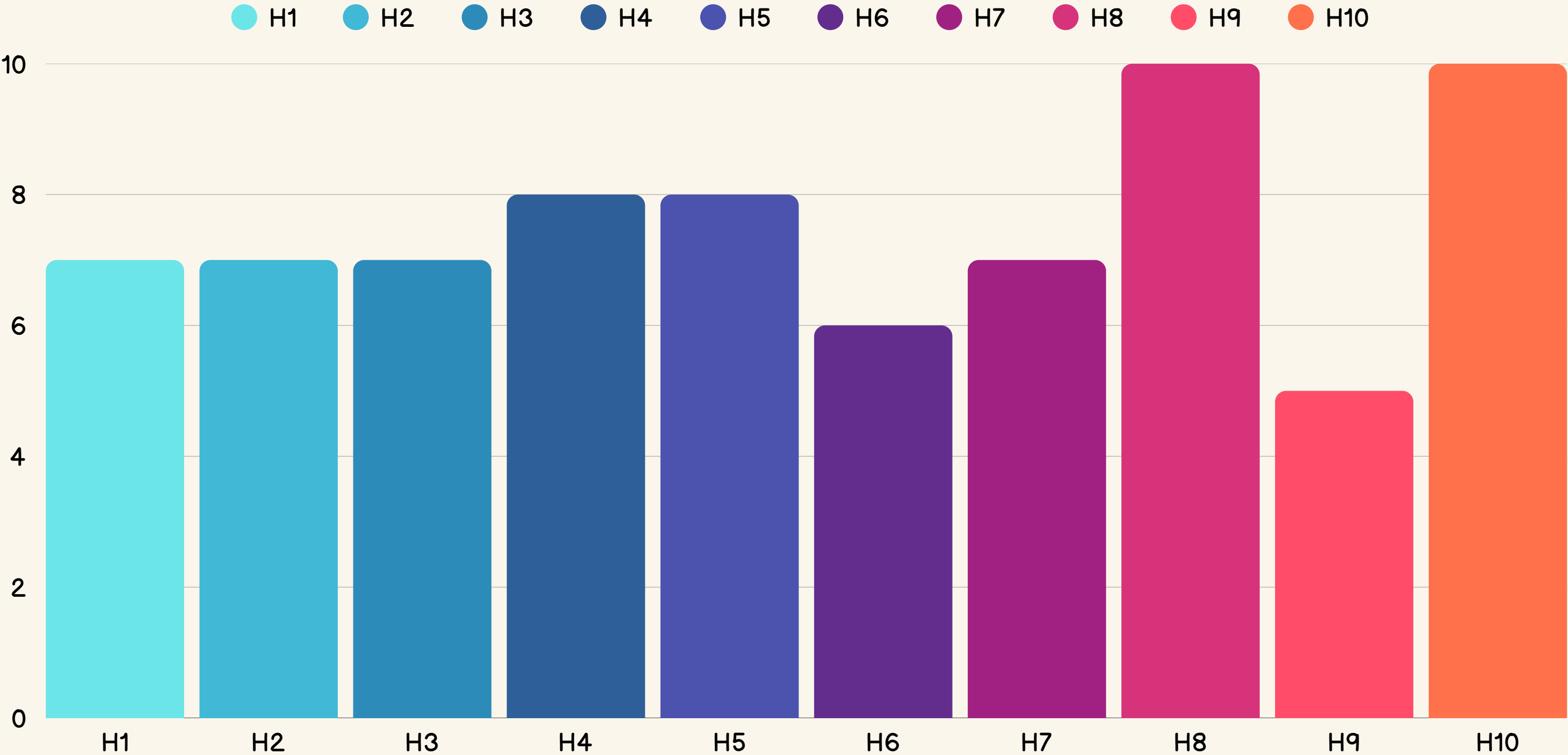
Most violations fall into Severity 2–3, suggesting that the prototype is generally usable but still contains several interaction issues that create friction in key task flows.

75 total heuristic violations identified across three peer evaluations.



VIOLATIONS BY HEURISTIC

TOTAL = 75

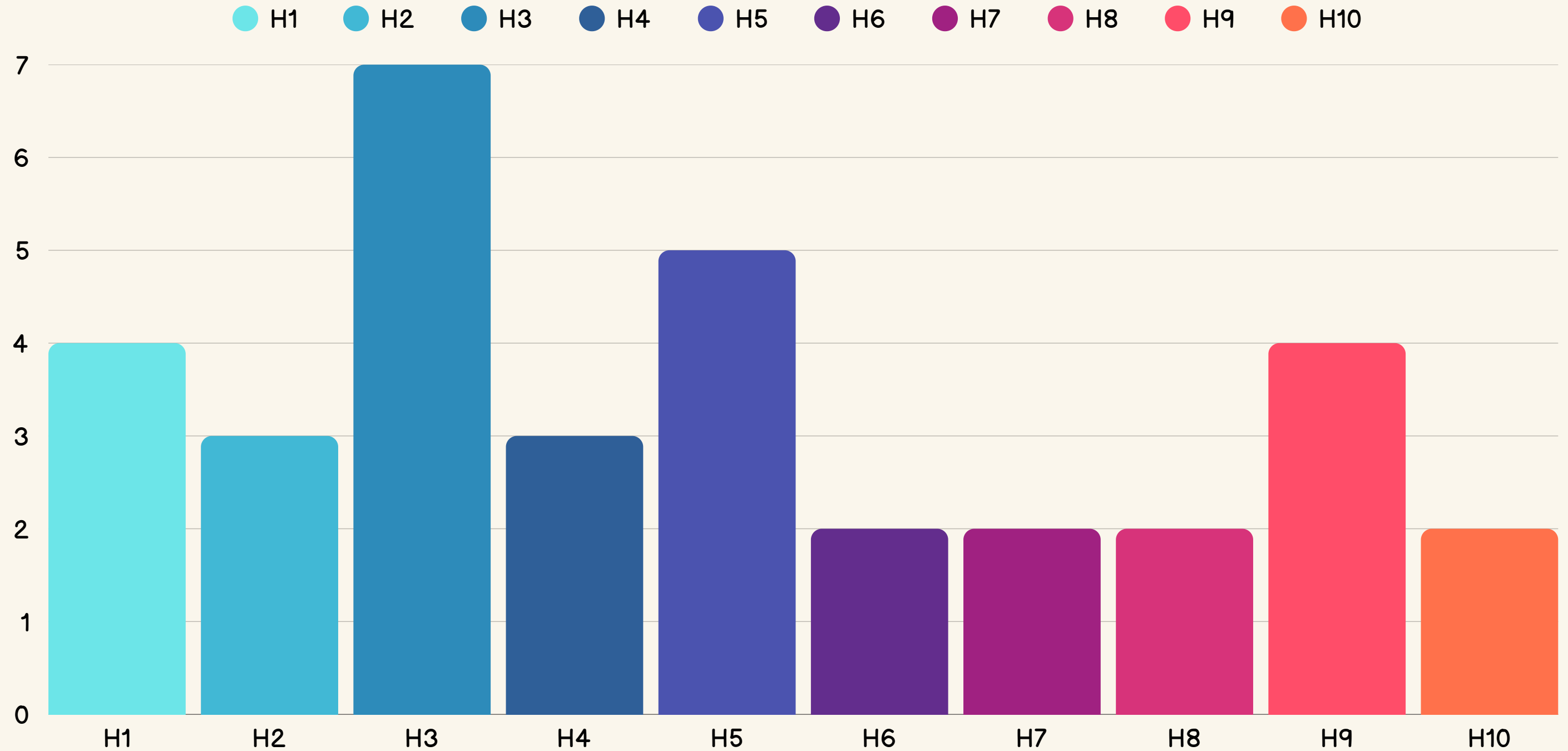


Most common violations include H8 (Aesthetic and Minimalist Design) and H10 (Help and Documentation), followed by H4 (Consistency) and H5 (Error Prevention), suggesting that users may experience visual clutter, inconsistent interaction patterns, and limited guidance while using the prototype.

GR8

HIGH LEVEL VIOLATIONS BY HEURISTIC (LV3,4)

TOTAL CHANGES TO
BE MADE =34



Most high-severity violations occur in H3 (User Control), H5 (Error Prevention), H9 (Error Recovery System) and H1 (System Status Visibility), indicating that users may experience friction when trying to maintain control, avoid errors, working around errors, and understand system feedback during key interactions.

SUMMARY OF VIOLATIONS

- User control and recovery limitations (H3)
- Weak error prevention mechanisms (H5)
- Limited visibility of system status and progress feedback (H1)
- Limited guidance for recognizing and recovering from errors (H9)
- Mismatch between system terminology and users' real-world understanding (H2)
- Inconsistent UI patterns and navigation behaviors (H4)
- Interfaces rely on recall rather than recognition (H6)
- Limited efficiency and flexibility for quick interaction (H7)
- Visual clutter and lack of minimalist design (H8)
- Insufficient onboarding, help, and documentation (H10)

The most critical high level issues appeared in User Control & Freedom (H3) and Error Prevention (H5), followed by System Status Visibility (H1) and Error Recovery Support (H9).

PRIORITIZING OUR VALUES AND GOALS

- Inklusivity:

Climbing solutions should adapt to different bodies rather than assuming a single “correct” way to climb. The system supports diverse physiques, abilities, and constraints by generating personalized betas that reflect each climber’s reach, movement style, and preferences.

- User Agency

Climbers control how much assistance they receive, when to progress, and which strategy to follow. The AI supports decision-making by offering personalized beta suggestions, but the climber ultimately decides which path to take.

- Accessibility

The system minimizes barriers to participation by using devices climbers already own and workflows that fit naturally into gym environments. Interaction should remain quick, simple, and usable even during active climbing sessions.

- Clarity of System Feedback

Users should always understand what the system is doing and what will happen next. Clear progress indicators, loading feedback, and confirmation messages help climbers quickly interpret results without interrupting their flow.

- Low Cognitive Load in a Climbing Context

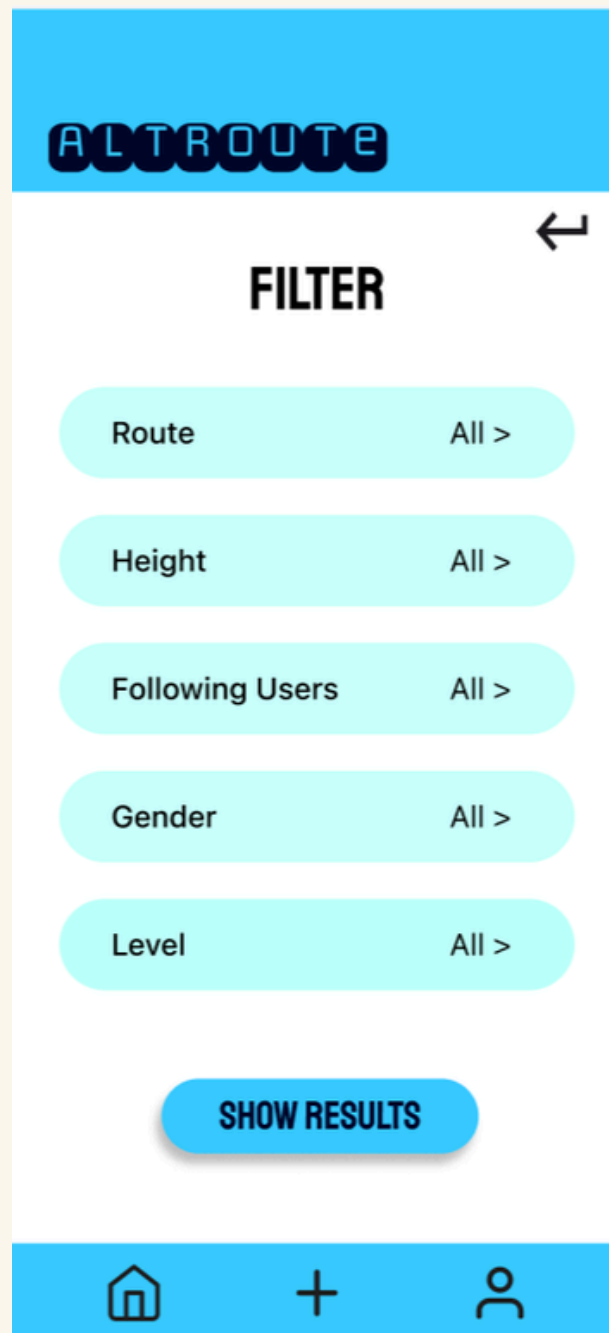
Because climbing requires physical focus and quick decision-making, the interface should minimize mental effort. Information should appear in a clear, glanceable format so climbers can understand beta steps quickly without unnecessary interaction.

UI & PRODUCT REVISIONS



USER CONTROL AND RECOVERY LIMITATIONS (H3)

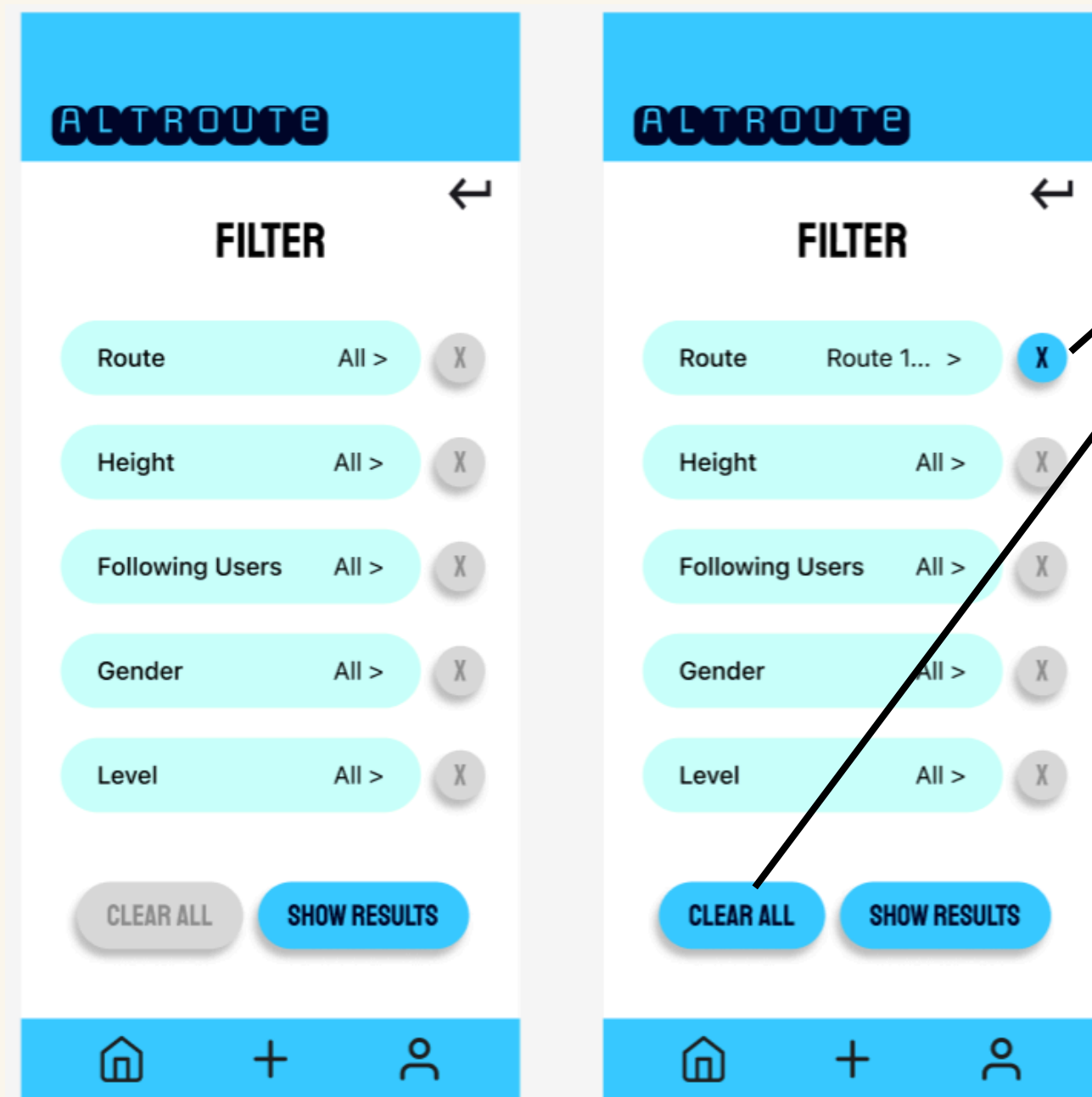
BEFORE



On Filter screen, there's no obvious "Clear all" / "Reset filters" escape hatch.

Users can get stuck in a narrow filter state

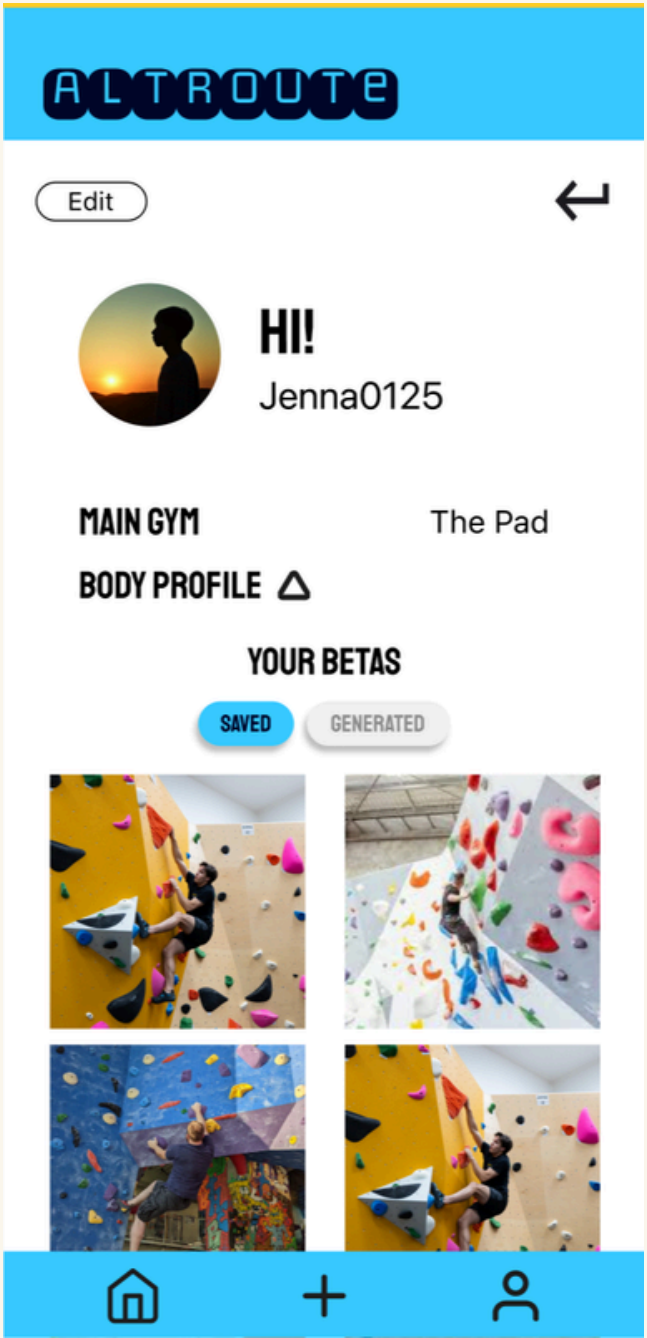
AFTER



Added "Clear All" button

USER CONTROL AND RECOVERY LIMITATIONS (H3)

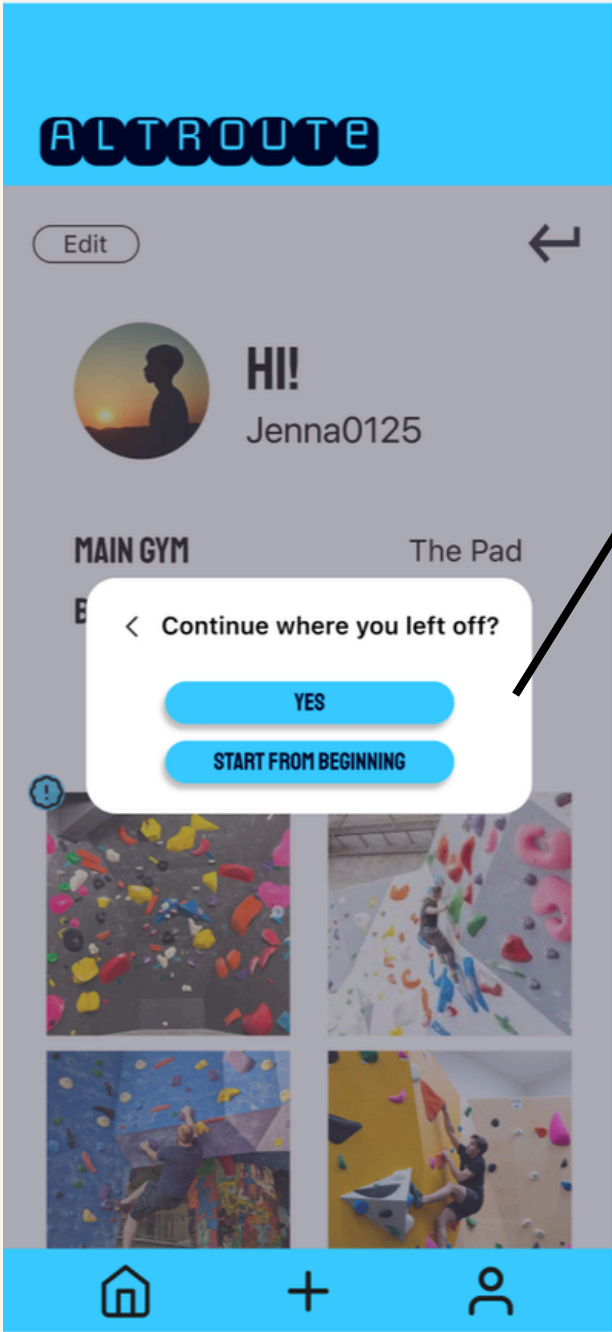
BEFORE



Users are not able to resume the generated beta from where they left off.

In the middle of climbing, it can increase confusion

AFTER



New page: Do you want to continue where you left off? / (Start from beginning, Continue)

USER CONTROL AND RECOVERY LIMITATIONS (H3)

BEFORE



When going through AI Beta instructions, there's no way to go back to a certain instruction once you've clicked, even if you use the back button.

Users can miss an instruction or accidentally click the screen to move forward and not be able to go back and read that step again.

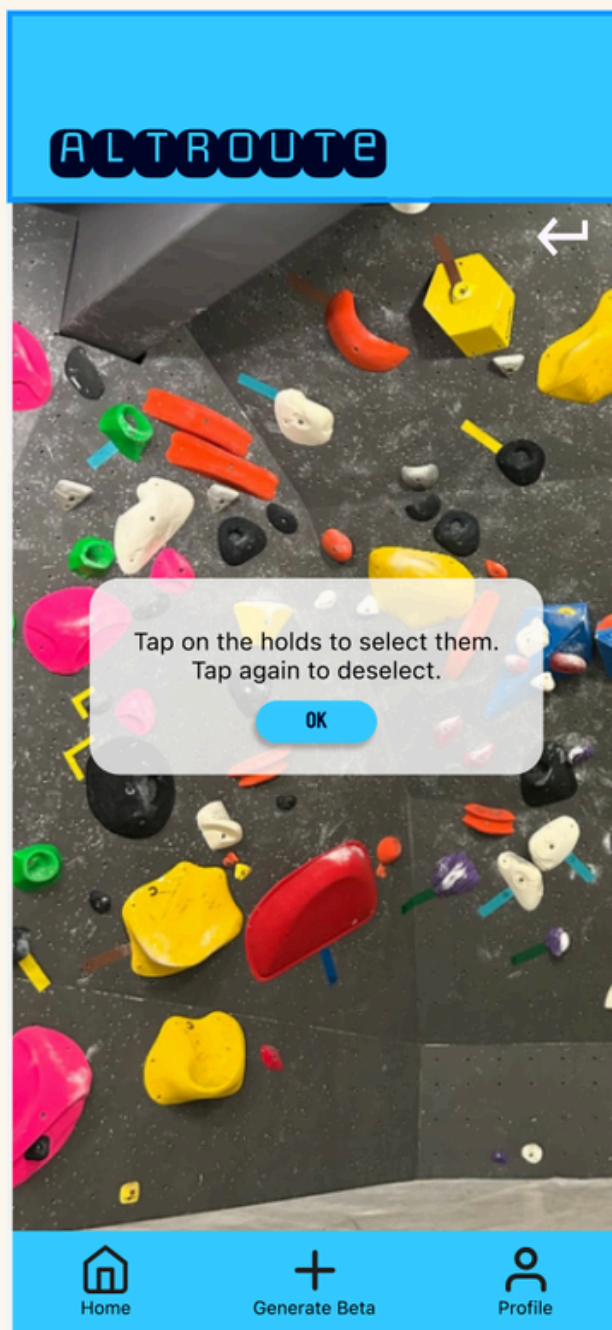
AFTER



Added going backward and forward buttons

USER CONTROL AND RECOVERY LIMITATIONS (H3)

BEFORE



- Once you've selected a hold, if you go back, it makes you take a photo again instead of being allowed to select another item.
- Hold selection step: users may mis-tap holds, but it's not obvious there is an easy undo/reselect mechanism (and slides mention selection order limitations).

AFTER



Users are supposed to tap on the holds again to deselect. We added this information to all hold-selection screens to help users if they meet with such errors. Tapping on that icon will show the same modal to remind users how to select and deselect holds.

USER CONTROL AND RECOVERY LIMITATIONS (H3)

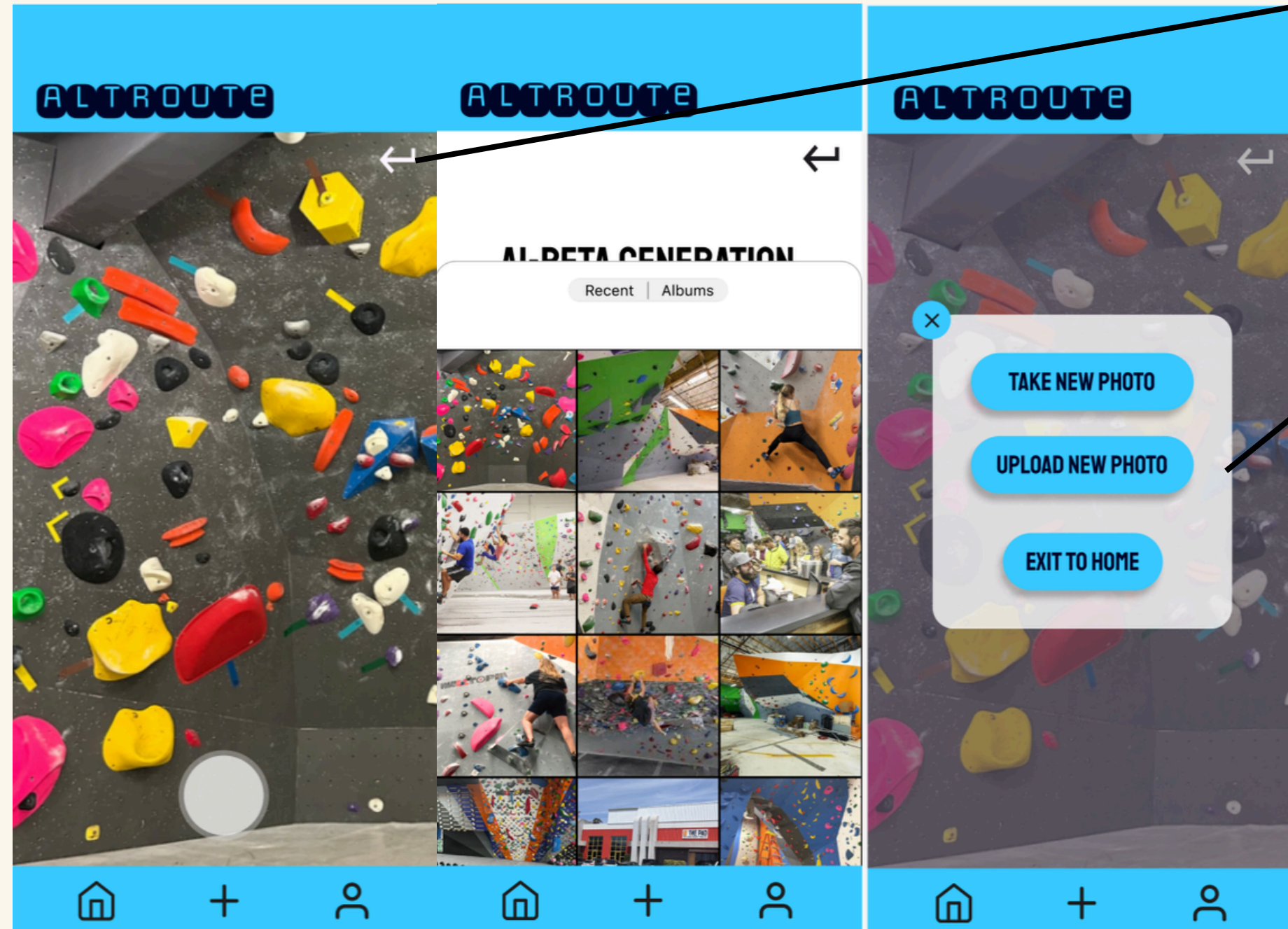
BEFORE



Once users upload a photo, there's no option to undo or select another photo without restarting the process over again.

Users may misclick and select the wrong photo, in which they would have to restart the whole progress instead of being able to simply select a new photo.

AFTER



Provided multiple options through back button (take new photo, upload photo, exit to home).

Users can now reupload a new route when selecting holds for the route they initially uploaded.

LIMITED VISIBILITY OF SYSTEM STATUS AND PROGRESS FEEDBACK (H1), (H6)

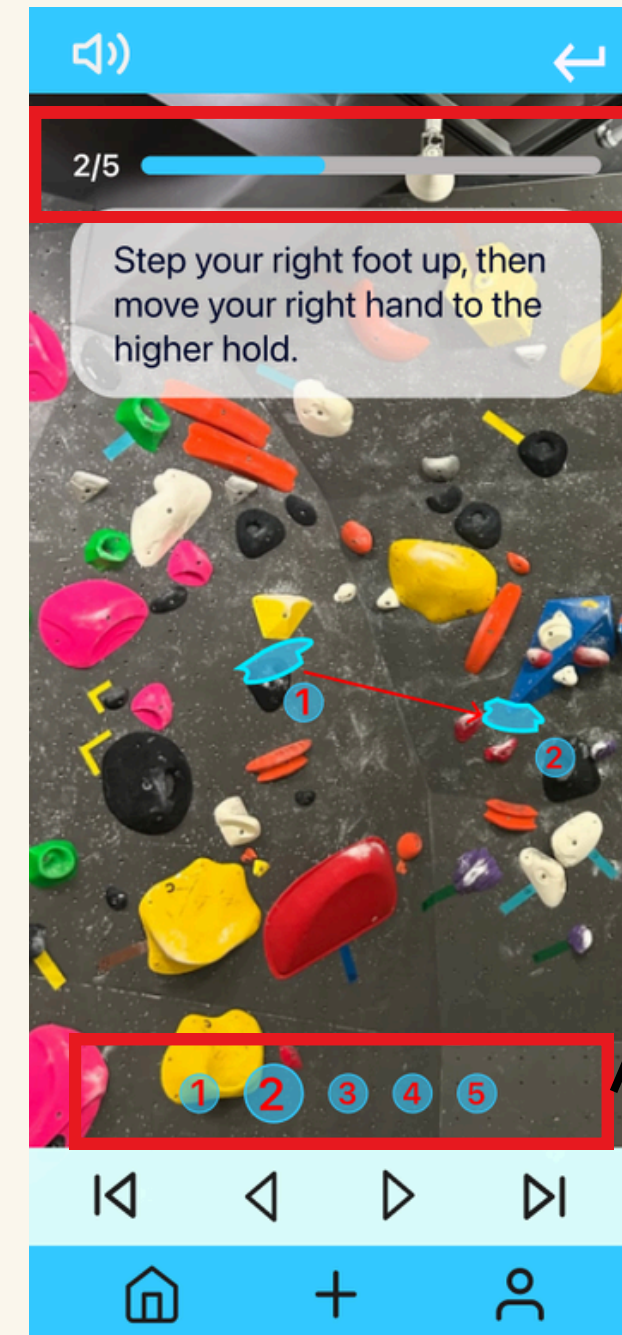
BEFORE



Step-by-step guidance relies on a progress indicator, but it's easy to lose track of "current step" if you leave the flow; status visibility is fragile.

During the climb uncertainty can cause disruption and safety problem

AFTER



Enlarged the previous progress bar and place it at the top for improved visibility.

Added selectable step indicator to quickly navigate through steps.

LIMITED VISIBILITY OF SYSTEM STATUS AND PROGRESS FEEDBACK (H1)

BEFORE

Generating your personalized beta...



Static loading bar

After clicking “Generate AI Beta,” there’s no clear loading/progress state indicating the system is “processing.”
Users may think it didn’t work.

AFTER

Generating your personalized beta...

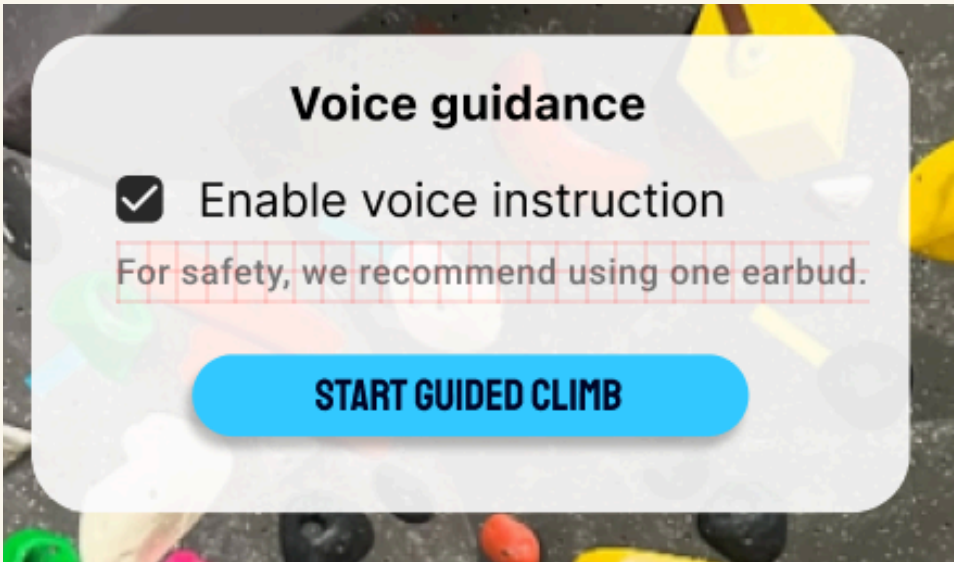


Animated loading bar

We animated the loading bar for a more obvious indication that the system is loading.

WEAK ERROR PREVENTION MECHANISMS (H5)

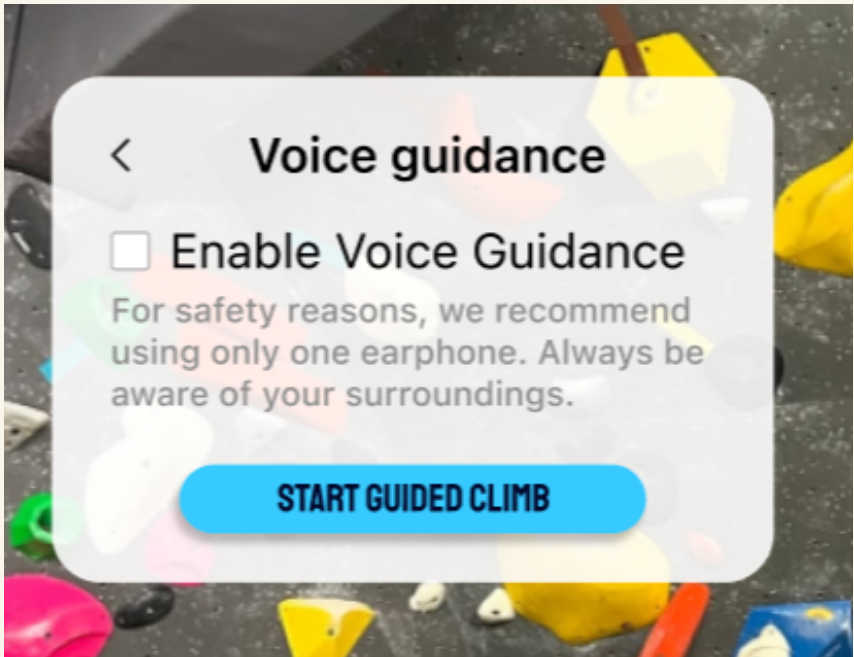
BEFORE



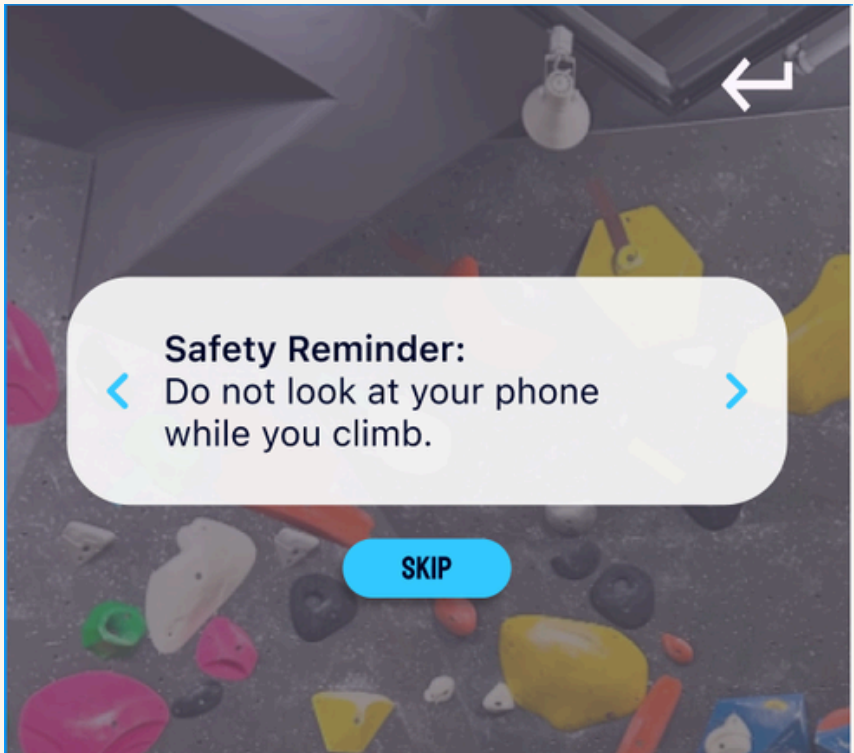
“Start Climbing” step mode encourages using the phone mid-climb, but there’s no safety friction

In a gym, distraction can be dangerous

AFTER



Added a short safety modal before starting the climb.



WEAK ERROR PREVENTION MECHANISMS (H5)

BEFORE

ALTRROUTE

BUILD YOUR BODY PROFILE

Height

5'7"

Arm Reach

5'9"

Skill

Top Level

v7

☐ Static Control

☐ Dynamic Movement

☐ Drop Knee

☐ Heel Hook

☐ Toe Hook

☐ Flagging

Current Injuries

☐ Shoulder Injury

☒ Finger Injury

☐ Sensitive Knee

GENERATE AI BETA

System appears to allow progression without clear required field validation.

Preventable input errors should be caught early.

AFTER

Basic Info*

Height

Arm Reach

Top Level

Skill

☐ Static Control

☐ Dynamic Movement

☐ Drop Knee

☐ Heel Hook

☐ Toe Hook

☐ Flagging

Current Injuries

☐ Shoulder Injury

☐ Finger Injury

☐ Sensitive Knee

GENERATE AI BETA

ALTRROUTE

BUILD YOUR BODY PROFILE

Basic Info*

Height

5'7"

Arm Reach

5'9"

Top Level

v7

Skill

☐ Static Control

☐ Dynamic Movement

☐ Drop Knee

☐ Heel Hook

☐ Toe Hook

☐ Flagging

Current Injuries

☐ Shoulder Injury

☐ Finger Injury

☐ Sensitive Knee

Added asterisk to indicate required information, grouped required information under "Basic Info"

The generate AI button is activated only after the required fields are filled

WEAK ERROR PREVENTION MECHANISMS (H5)

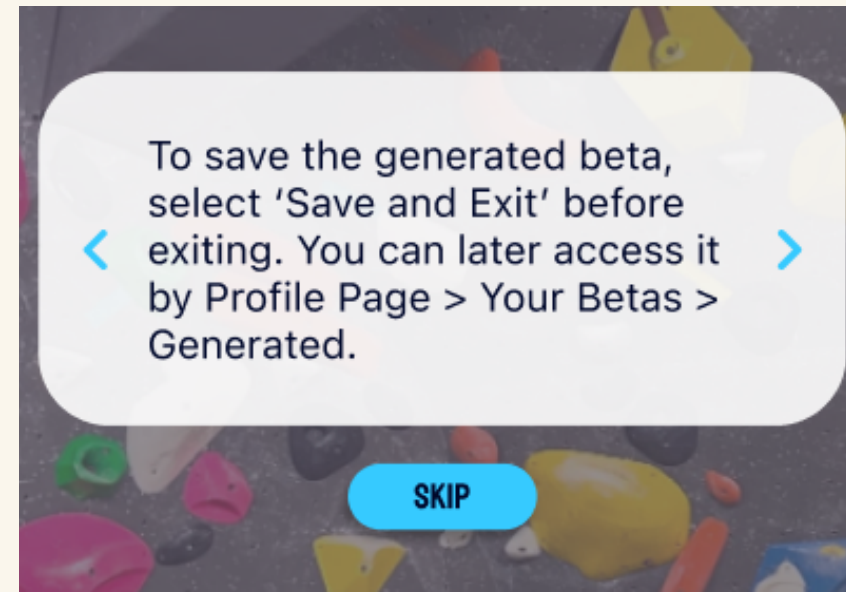
BEFORE



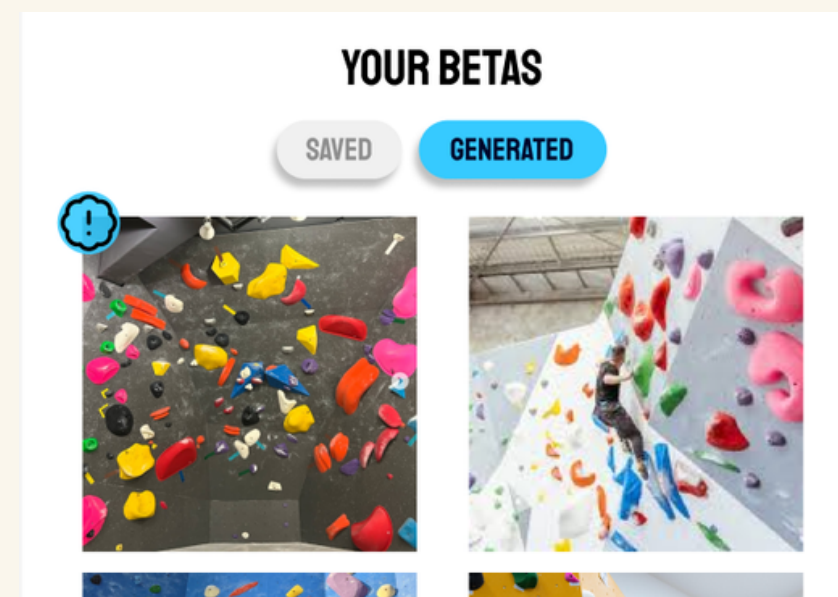
Generating a new beta may override the previous one.

Risk of losing information without confirmation.

AFTER



Added note to inform that betas are saved to Profile, and can be accessed from there.



Newly generated beta indicated by '!' icon on the top left corner.

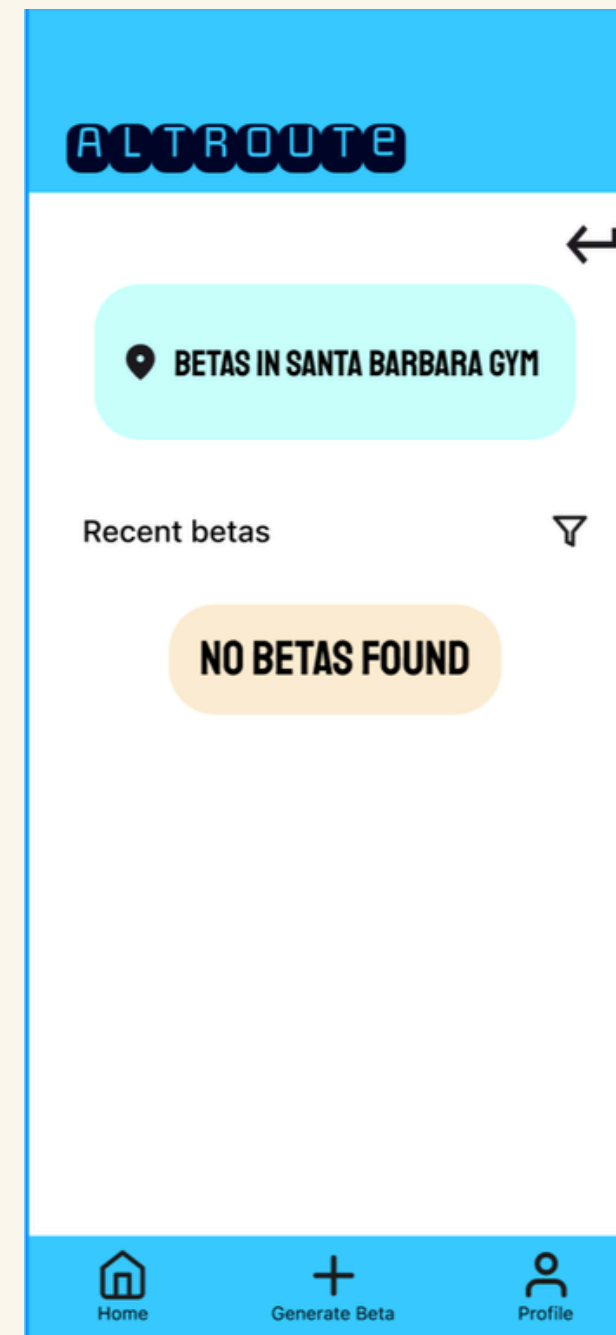
LIMITED GUIDANCE FOR RECOGNIZING AND RECOVERING FROM ERRORS (H9)

BEFORE

If filters return no results, there's no clear error state ("0 results") plus guidance.

Users don't know what went wrong or how to fix it.

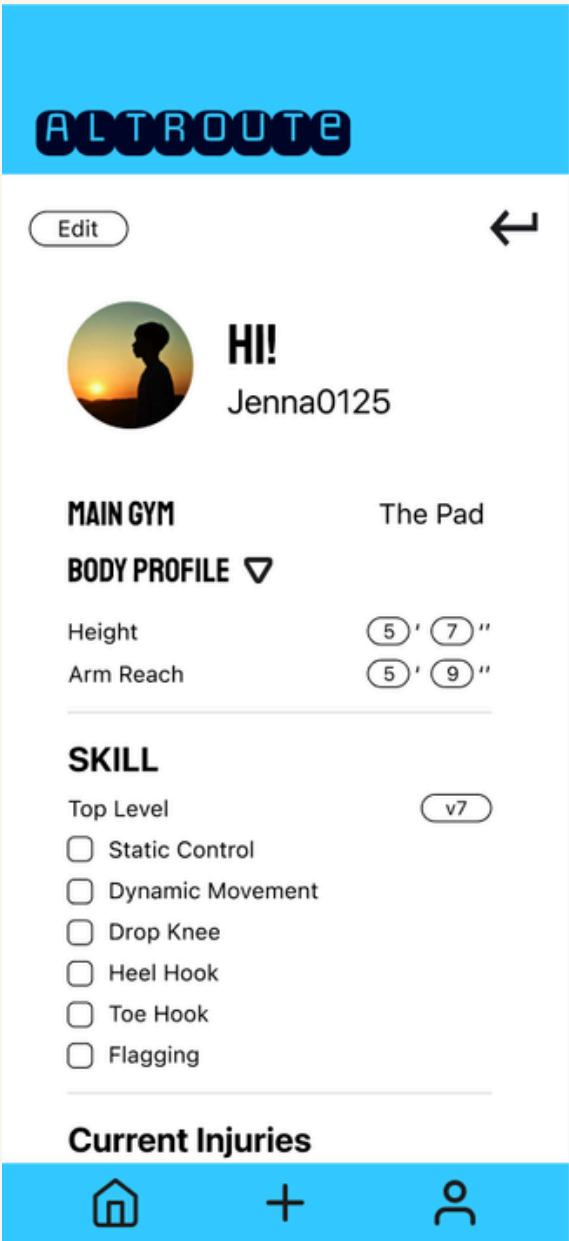
AFTER



Added no beta found notification

MISMATCH BETWEEN SYSTEM TERMINOLOGY AND USERS' REAL-WORLD UNDERSTANDING (H2)

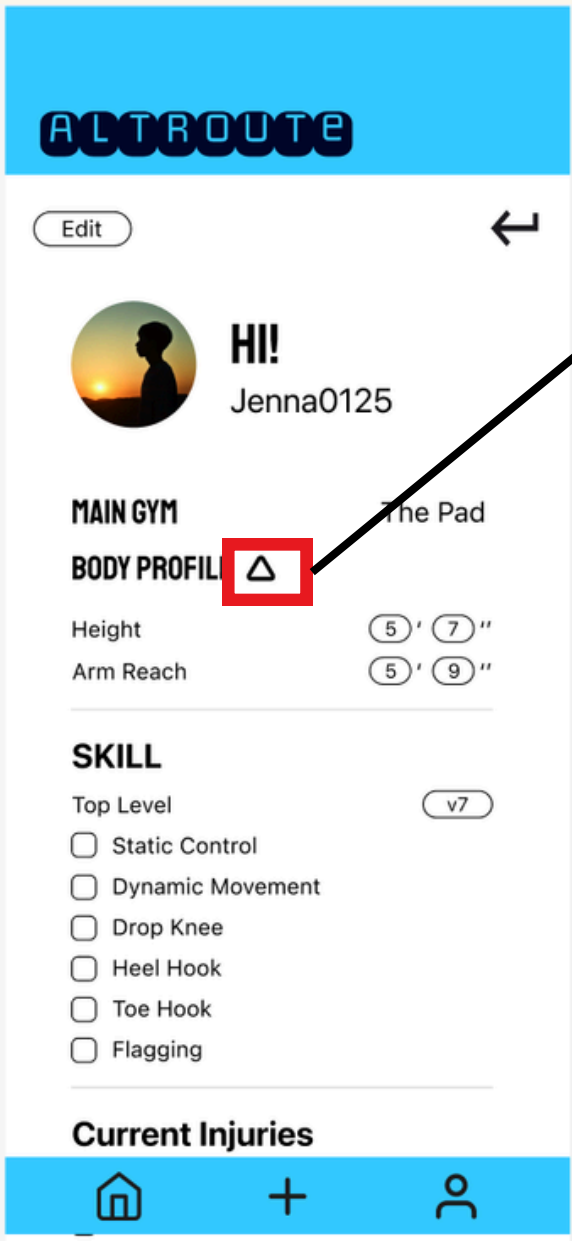
BEFORE



Body Profile drop down icon is mismatched with real world standards. The down arrow should be clicked when wanting to expand, and the up arrow should be clicked in order to show less, however it's the opposite in the prototype

In the real world, when users want to expand something, they should click the down arrow. However, in this case, it's swapped which would confuse users if the "body profile" is actually expanded or not.

AFTER



Fixed the icon

MISMATCH BETWEEN SYSTEM TERMINOLOGY AND USERS' REAL-WORLD UNDERSTANDING (H2)

BEFORE



Beta Video has play/pause signals mixed up. (Ex: When wanting to pause the video, the user has to click on a play button, but it should be the other way around)

It may confuse the user since in the real world, the pause button should be clicked when wanting to pause the video. Users may think it's a glitch.

AFTER



Swapped the play/pause buttons.

MISMATCH BETWEEN SYSTEM TERMINOLOGY AND USERS' REAL-WORLD UNDERSTANDING (H2)

BEFORE

ALTROUTE

←

BUILD YOUR BODY PROFILE

Height

'

"

Arm Reach

'

"

Skill

Top Level

☐ Static Control

☐ Dynamic Movement

☐ Drop Knee

☐ Heel Hook

☐ Toe Hook

☐ Flagging

Current Injuries

☐ Shoulder Injury

☐ Finger Injury

☐ Sensitive Knee

GENERATE AI BETA

Injury selection appears simplified and may not reflect real injury nuance (severity, side, limitations).

Real-world injuries are complex; oversimplification may create unrealistic expectations.

AFTER

Current Injuries

☐ Shoulder Injury

☐ Finger Injury

☐ Sensitive Knee

Additional Notes

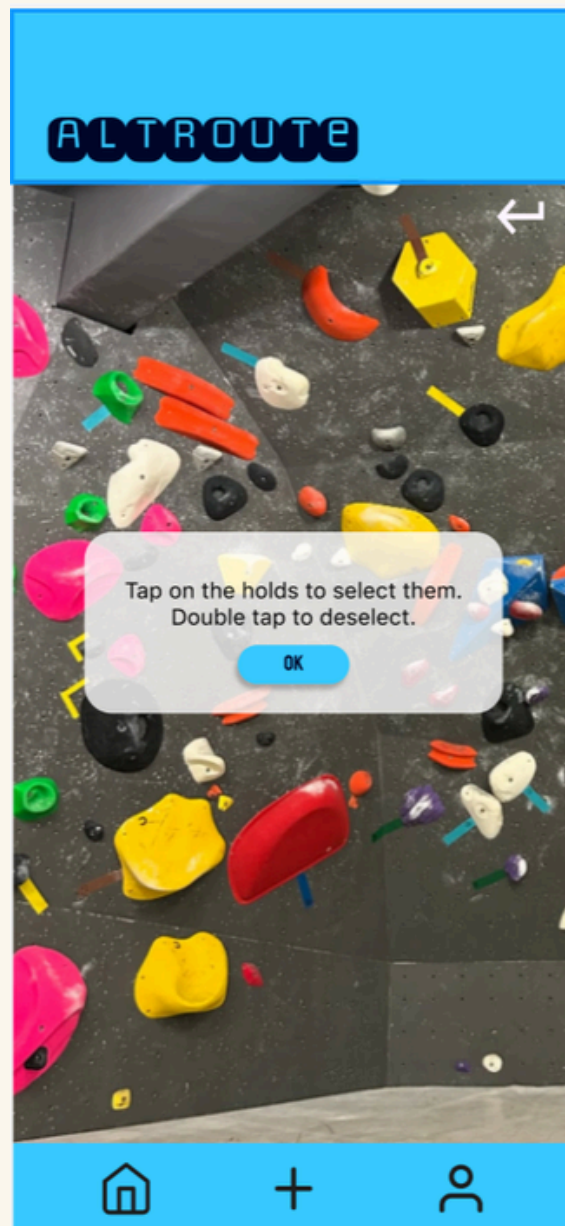
Input additional considerations...

GENERATE AI BETA

Added "additional information" section

INCONSISTENT UI PATTERNS AND NAVIGATION BEHAVIORS (H4)

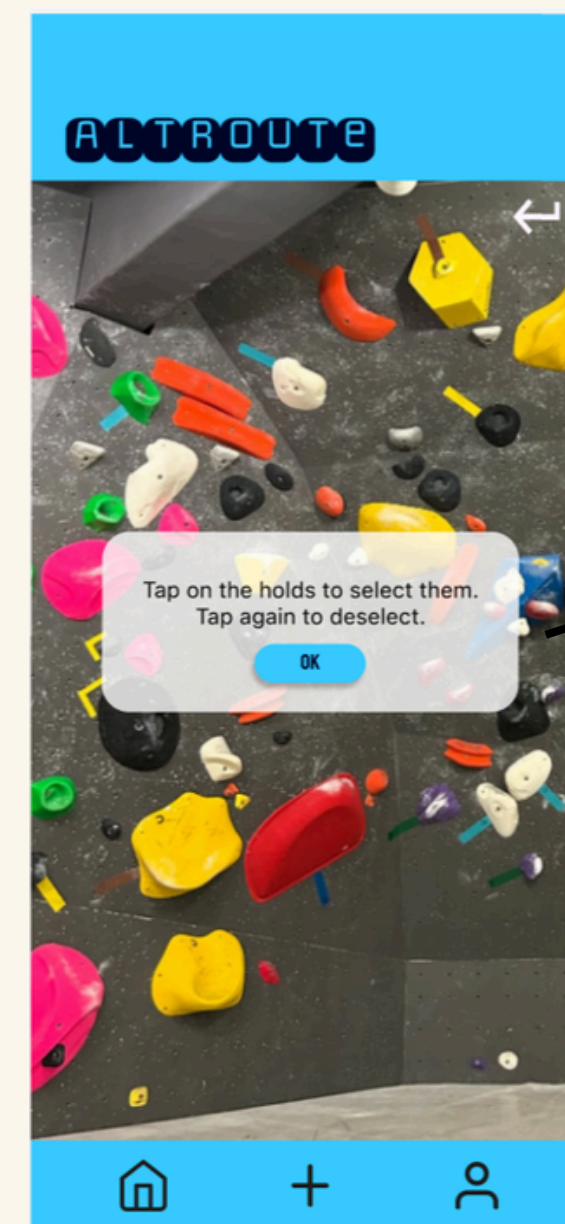
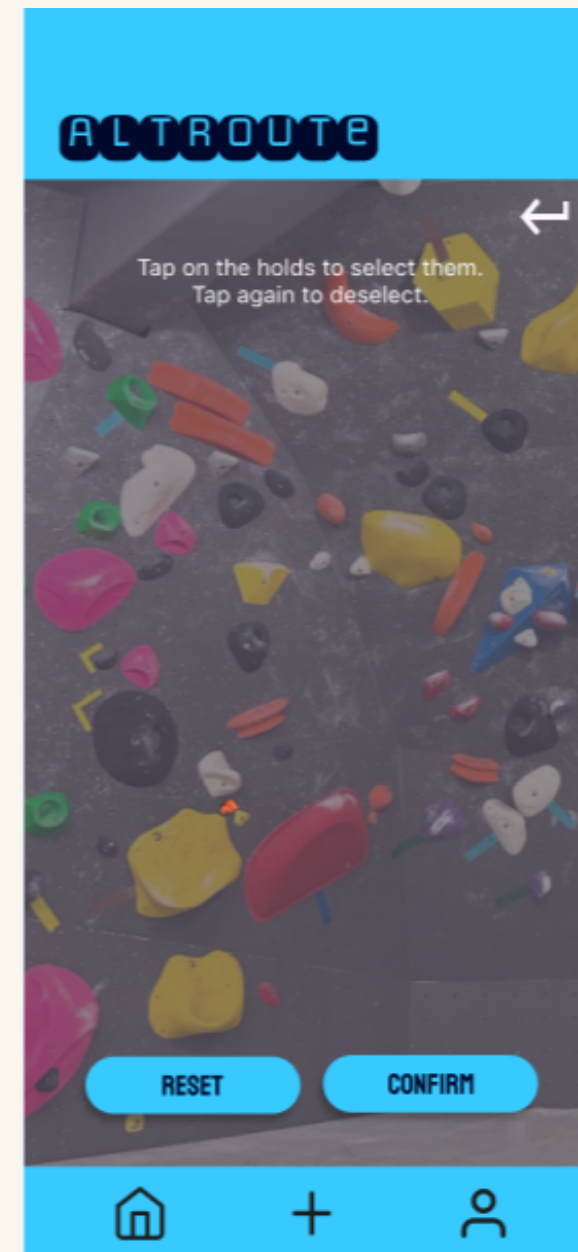
BEFORE



The instructions say Tap on the holds to select them. Tap again to deselect" when taking a photo but it changes to "Tap on the holds to select them. Double tap to deselect" when uploading a photo". In reality, the user will be tapping again to deselect, not double tapping.

Users will get confused if they have to "double tap" to deselect since in reality, they would be simply tapping again.

AFTER



Changed the text to "Tap on the holds to select them. Tap again to deselect" for both options.

INCONSISTENT UI PATTERNS AND NAVIGATION BEHAVIORS (H4)

BEFORE



After uploading an image, and clicking on a hold, if you click the back button, it makes you take a photo, rather than just going back to the original image that was uploaded.

That confuses the user since they selected the upload button and would want to go back to the original image.

AFTER



Fixed the user flow. The back button now asks users if they would like to reupload a photo or take a new photo.

LIMITED EFFICIENCY AND FLEXIBILITY FOR QUICK INTERACTION (H7)

BEFORE



When going through AI Betas, the instructions pop up everytime and there's no button to skip them in the case that you've seen them before.

For users who have used the app before or are familiar with the instructions, it could be frustrating to keep having to click through them whenever they generate a new beta.

AFTER



We added skip button and made it possible to skip the instruction

LIMITED EFFICIENCY AND FLEXIBILITY FOR QUICK INTERACTION (H7)

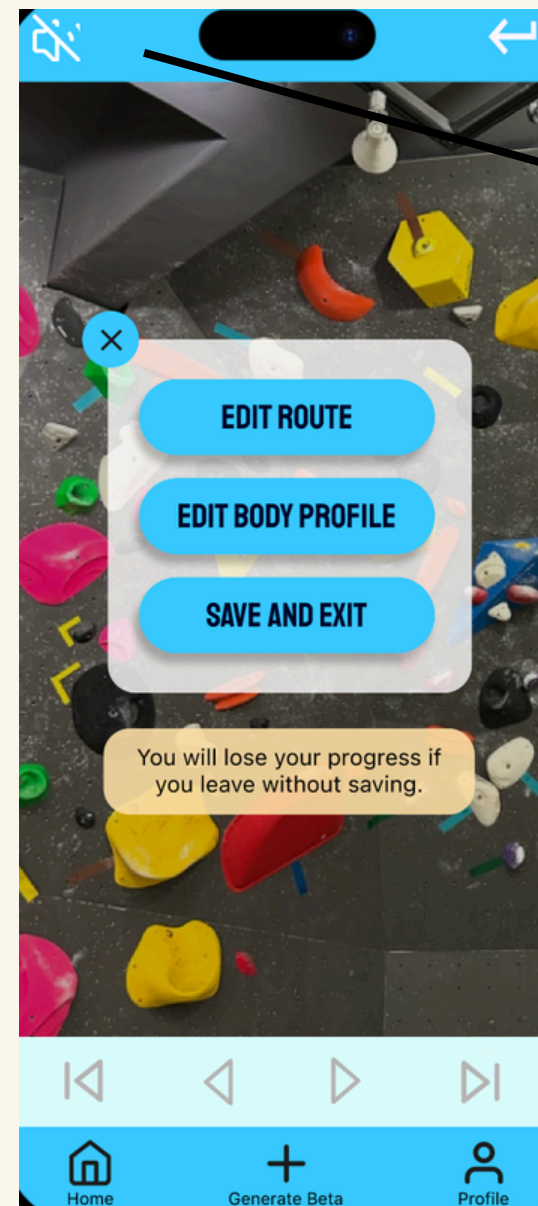
BEFORE



Voice is optional but unclear if switchable mid-session.

Climbers may change preference mid-climb.

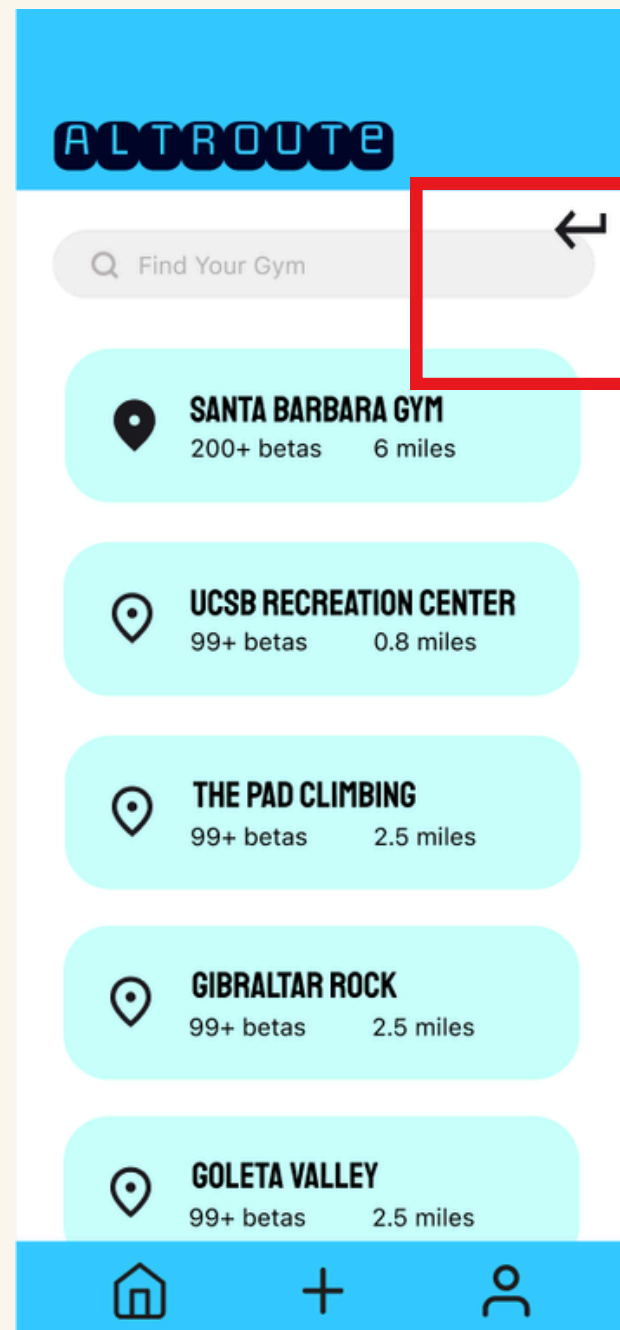
AFTER



Able to turn on and off voice mode

VISUAL CLUTTER AND LACK OF MINIMALIST DESIGN (H8)

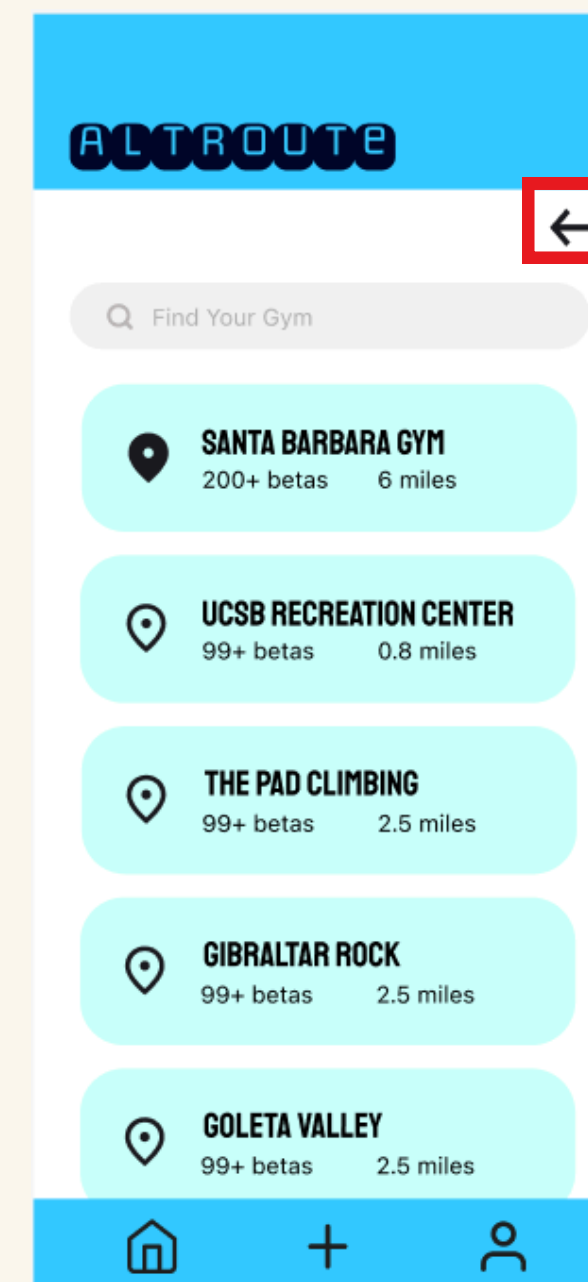
BEFORE



Part of the back button overlaps with the "Find your Gym" tab.

Users can misclick on the button when they mean to click another.

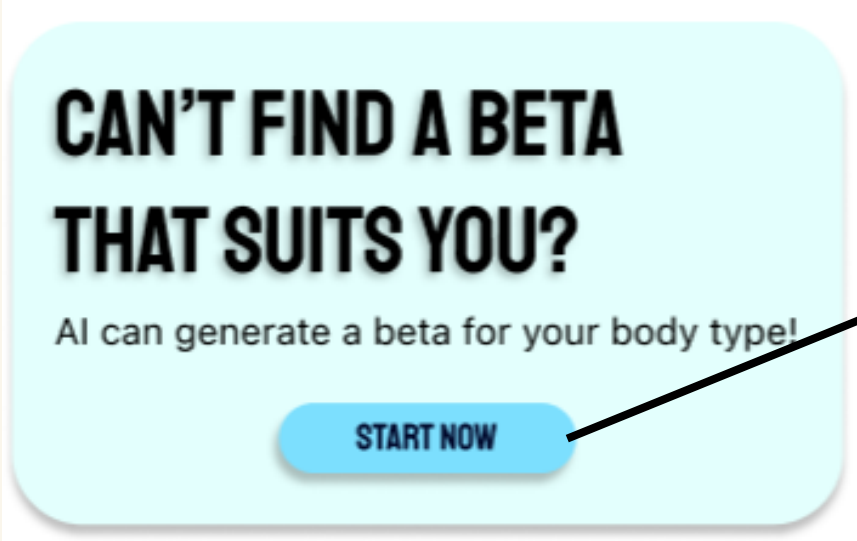
AFTER



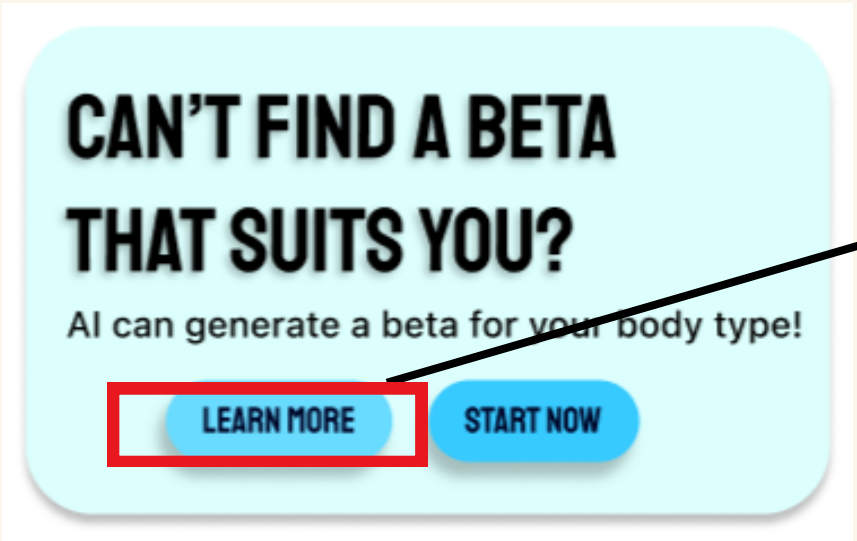
Fixed the Margins

INSUFFICIENT HELP, AND DOCUMENTATION (H10), (H3)

BEFORE



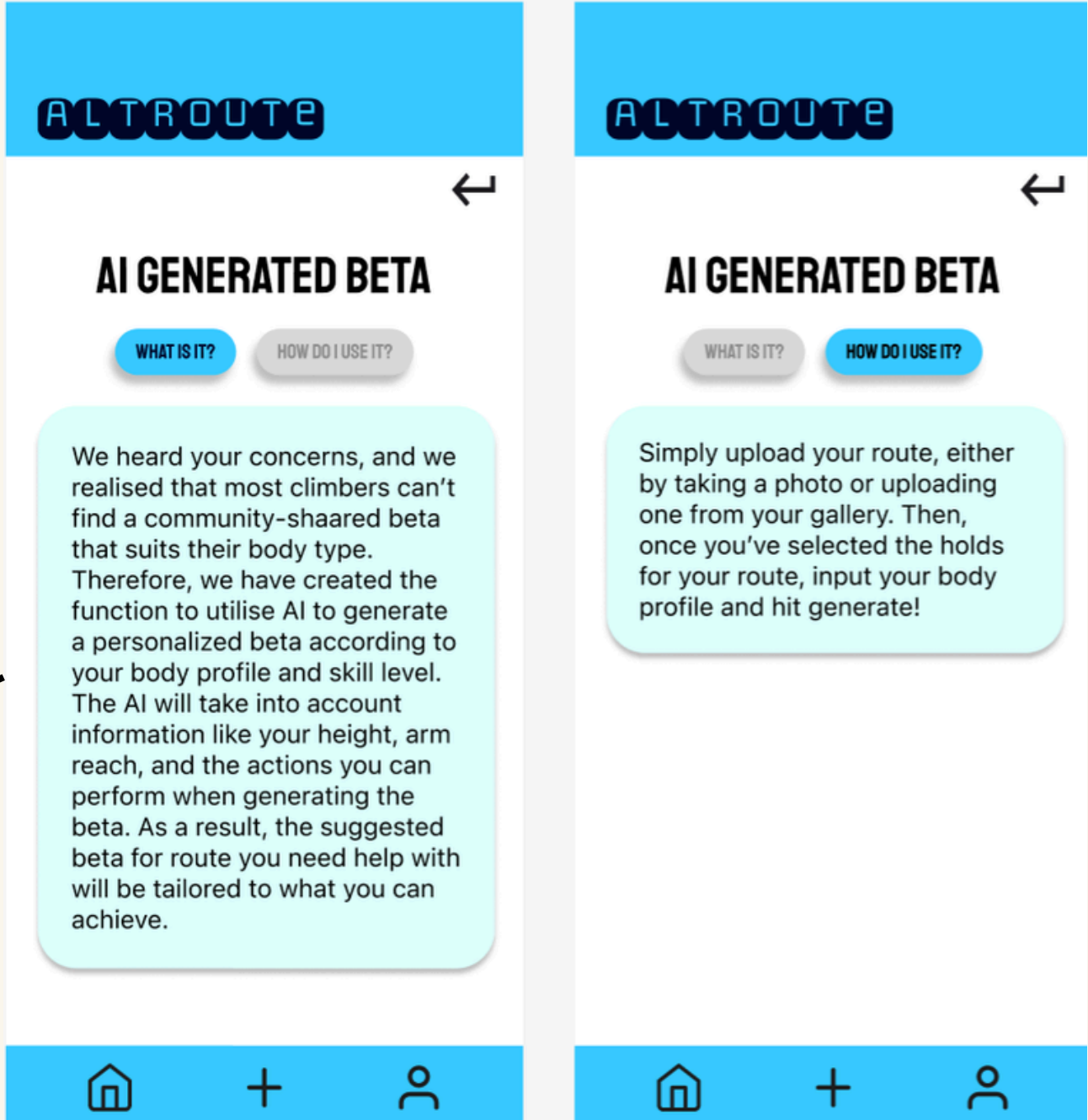
AFTER



Users confused about AI-generated vs custom beta
Complex mental model requires explanation.

Learn more documentation added

AFTER



FIXED HEURISTICS

- H1 – Visibility of System Status: Added clearer progress indicators and a loading animation so users can better understand when the system is generating AI betas.
- H3 – User Control & Freedom: Added features such as reset filters, navigation controls, and a resume option so users can easily recover from actions and move.
- H5 – Error Prevention: Required fields and a safety prompt were added before starting the climbing guidance to help prevent common user mistakes.
- H2 & H4 – Real-World Match and Consistency: Fixed several UI elements such as icons, button behavior, and interaction instructions.
- H6 & H7 – Recognition and Efficiency: We added a preview of the last session and a skip option so users don't have to repeat instructions they have already seen.
- H8 & H10 – Minimalism and Help: We cleaned up parts of the interface to reduce visual clutter and added help section to explain how AI-generated betas work.

DISAGREED HEURISTICS SUGGESTIONS

- Users can easily submit incomplete/invalid profiles and blame the app when results feel wrong. (H5)
> In our original prototype, when there is no input, button will not be activated, cannot proceed.
- If users select holds incorrectly, there's no error feedback ("Selected holds don't form a route") or recovery suggestion. (H9)
> System wouldn't know what's right or wrong, and it could predict wrongly as well.
- After clicking "Generate AI Beta," there is no clear loading state or explanation of what the system is doing. (H1)
> Made loading state obvious with animated loading bar. Users are not meant to know how the AI generates their beta.
- Although progress indicators were added, it's unclear how many total steps remain at a glance. (H1)
> Total steps remaining were already implemented.
- Hold selection only works in specific order due to Figma constraints. (H3)
> The prototype presents how users should select holds, although not necessarily in that specified order. We were not able to implement this level of interactivity on Figma.

DISAGREED HEURISTICS SUGGESTIONS

- Exiting step-by-step mode may reset flow.(H3)
>Exiting actually saves the generated beta to the users Profile page, but it wasn't made clear in the previous prototype. This has been improved through the instructions provided, as mentioned on slide 29.
- Resume feature is described but not implemented in prototype.(H6)
>Duplicate suggestion from another peer feedback.
- Height, reach, skill, injuries appear clustered.(H8)
>We did not agree that it appeared clustered, but to improve it, we increased the spacing.
- Empty state handling unclear.(H9)
>Duplicate suggestion from another peer feedback.
- Body profile does not affect output.(H9)
>Figma prototype explains the user flow clearly.
- After users enter height, reach, skill level, and injuries, the system does not explain how those inputs are used to generate the personalized beta.(H10)
>We do not agree that users need to know how AI uses their information to generate their betas.

IMPACT OF PROGRESS TOWARDS GOALS



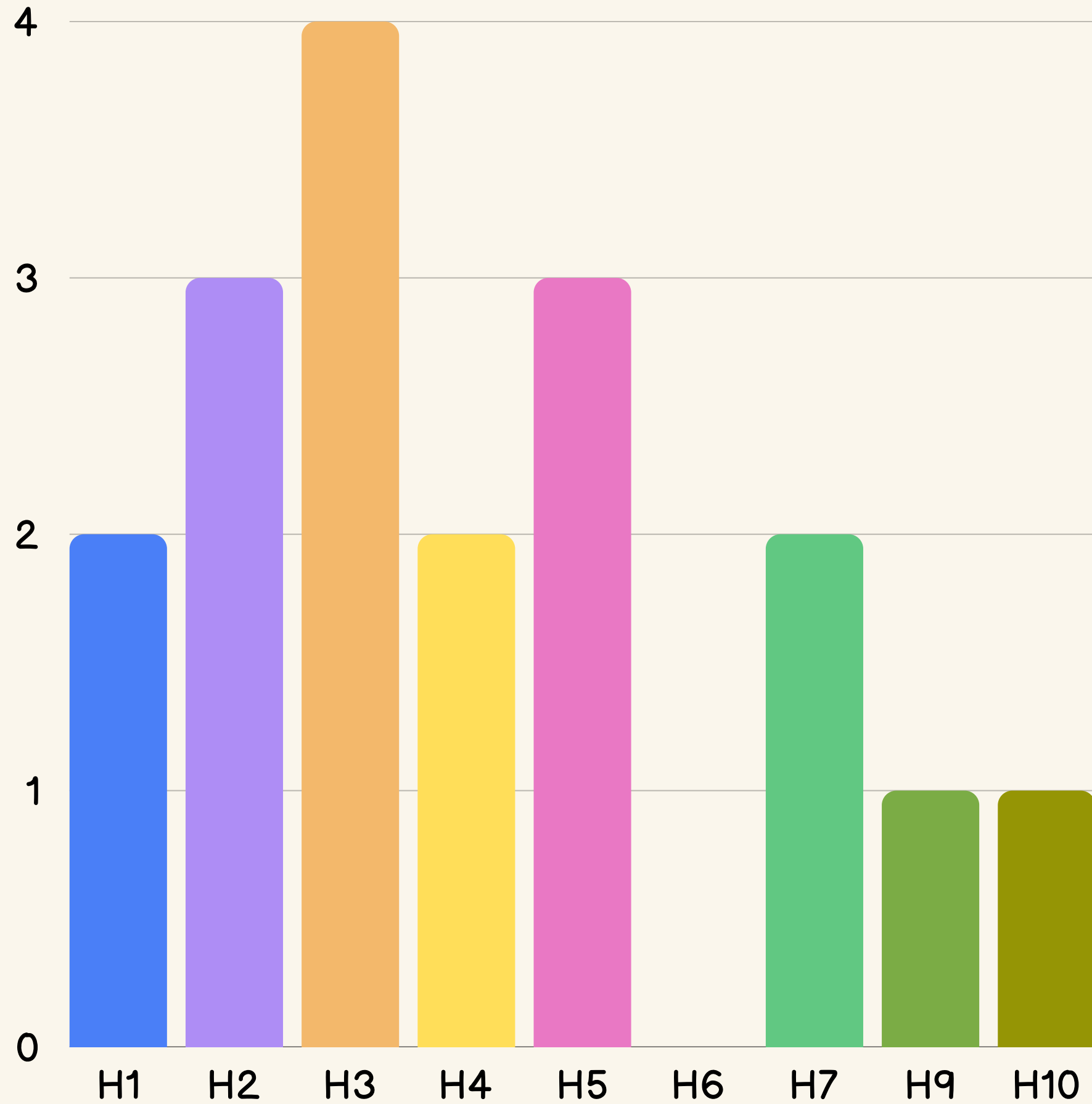
GR8

FIXED HEURISTICS

H1~H10 : 19

Rejected or Overlapping changes: 15

Total = 34



OUR USABILITY GOALS

Efficiency

Clarity / Error Prevention

From previous testing results, users were able to complete all tasks efficiently, meeting our first goal. However, clarity issues in the moderate (generate AI beta) and complex (use AI beta to climb) tasks prevent the design from fully meeting our usability goals.

With the heuristic evaluation, we implemented more bite-sized instructions to guide users on how to use the features to complete the tasks. We also ensured that users are able to go back and correct their mistakes if they had made errors during route and body profile inputting. These changes work towards the clarity and error prevention usability goal.

IMPACT OF PROGRESS TOWARDS GOALS

- Improved user control and recovery during climbing flows and beta generation
- Reduced user errors through clearer validation and safety prompts
- Increased visibility of system status during AI beta generation
- Reduced cognitive load for climbers in a real gym environment

PROTOTYPE IMPLEMENTATION STATUS



TOOL: FIGMA

Pros:

- Easy to iterate quickly after feedback
- Simple to connect screens and create multiple flows
- Works well for mobile layout and spacing
- No other installation needed

Cons:

- No real backend logic or AI processing
- Personalization is simulated
- Limited realism for video/voice interactions
- Designed not real data
- Hard to represent real-time system responses

IMPLEMENTED FEATURES

- Community Beta Search and Viewing

Users can search and watch beta videos uploaded by other climbers to learn different strategies.

- Route Filtering

Users can apply filters to narrow down routes based on relevant criteria.

- AI Beta Generation

The system generates personalized beta suggestions based on user inputs and route information.

- Step-by-Step Beta Guidance

Generated betas are presented as sequential climbing steps to guide movement and decision making.

- Saved Beta Library

Users can save generated betas and revisit them later.

- Gym Discovery

Users can search for climbing gyms and explore routes available at those locations.

- Help and Guidance Section

Users can access help documentation to better understand how to use the app.

UNIMPLEMENTED FEATURES

- Voice Interaction for Hands-Free Control

Allow climbers to navigate beta steps and control the interface using voice commands while climbing.

- AI Explanation for Generated Betas

Provide explanations that show why specific beta steps were suggested based on the user's body profile and route constraints

.

- Beta Upload and Sharing

Allow climbers to upload their own beta videos and share strategies with the community.

- Personalized Route Recommendations

Suggest routes and betas based on the user's climbing history, preferences, and skill level.

WIZARD OF OZ + HARD CODED FEATURES

- AI beta generation is pre-designed (output does not change)
- Body profile fields do not affect the result (result is fixed)
- Gym list and betas are sample data
- Save / Comment actions are visual only
- Step-by-step guidance is pre-written, not dynamically generated
- Voice guidance is simulated and not interactive

LINK TO DELIVERABLES

FIGMA:

[HTTPS://WWW.FIGMA.COM/DESIGN/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?NODE-ID=0-1&P=F&T=UGCGY1O2AXOKZABH-0](https://www.figma.com/design/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?node-id=0-1&p=f&t=UGCGY1O2AXOKZABH-0)

FIGMA PROTOTYPE:

[HTTPS://WWW.FIGMA.COM/PROTO/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?NODE-ID=0-1&T=CCM0V3KG4W2YWC03-1](https://www.figma.com/proto/MM2RDZLNWCBYRYNOD0TPXQ/ALTROUTE?node-id=0-1&t=CCM0V3KG4W2YWC03-1)

WEB:

[HTTPS://AAASAMI.GITHUB.IO/AITROUTE/#](https://aaasami.github.io/aitroute/#)

APPENDIX



CLIMBING TERMS

ROUTE/PROBLEM
A SPECIFIC
CLIMBING PATH SET
ON THE WALL
USING DESIGNATED
COLORED HOLDS.



SEND
A "SEND" IS ACHIEVED WHEN A
CLIMBER FINISHES A CLIMB
WITHOUT FALLING,
REGARDLESS OF WHETHER IT IS
THEIR FIRST TRY OR THEY HAVE
PRACTICED IT BEFORE.

BETA
A CLIMBER'S STRATEGY OR
SEQUENCE OF MOVES FOR
FINISHING A ROUTE.